

**Identifying and Redesigning Flow Process Chart in Real-Life
Processes: Laundry at Home**

OPMT 620: Operations Management (HBD- FALL25-05)

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Part 1: Current-State Flow Process Chart

Location: Home (Laundry Room)					Summary					
Activity: Washing and drying laundry process					Event	Present	Proposed	Savings		
Date: 18/11/2025					Operation	5	4	1		
Operator: Household member		Analyst: Process analyst			Transport	5	5	0		
Circle appropriate method and type: Method: Present Proposed Type: Worker Material Machine					Delay	1	1	0		
					Inspection	1	1	0		
					Storage	0	0	0		
Remarks: By reorganizing the laundry room and removing movement that’s not essential, time and effort can be significantly reduced.					Time (min)	138	75	63		
					Distance (ft)	670	305	365		
					Cost	100	70	30		
Event Description		Symbol			Time (min)	Distance (ft)	Method recommendation			
Collect dirty clothes from the rooms		○	→	D	□	▽	10	150	By placing laundry basket in each room, walking and any other unnecessary trips can be eliminated	
Sort the clothes based on color and fabric		○	→	D	□	▽	10	0	Place separate hampers to sort clothes to remove manual sorting	
Inspect for stains and treat it accordingly		○	→	D	□	▽	5	0	Merge sorting with stain inspection to reduce inspection time	
Transport the sorted clothes to the laundry room		○	→	D	□	▽	5	150	Use laundry hampers with wheel to minimize physical effort	
Load the washer and add detergent		○	→	D	□	▽	5	10	Use detergents that are measure beforehand like pods to enhance	

								consistency and reduce time measuring the detergent
Wait for wash cycle to complete	○	→	D	□	▽	45	0	Set a timer using a phone to alert when the cycle is complete instead of checking every now and then
Transfer wet clothes from the washer to the dryer	○	→	D	□	▽	5	40	Set the washer and dryer side by side to reduce time and physical effort
Start and dry the clothes in the machine	○	→	D	□	▽	30	0	Use a timer to alert when cycle is complete and use a high efficiency setting to finish faster
Remove from dryer and move to folding station	○	→	D	□	▽	6	70	Set up a folding station near the laundry room to reduce transportation time
Fold the clothes	○	→	D	□	▽	7	0	Dedicate a surface(table) to improve ergonomics
Move the folded clothes to their rooms	○	→	D	□	▽	5	150	To reduce trips to different rooms, group the folded clothes based on a person/room
Store them in drawers/closets	○	→	D	□	▽	5	100	Reorganize the storage area for frequently used items to increase efficiency

Part 2: Waste and Inefficiency Analysis

An analysis of the present-day laundry washing and drying process shows that there are multiple inefficiencies resulting from unnecessary movement, delays, and manual activities. While the workflow properly carries out the necessary work, three main areas of waste namely, needless transportation, waiting, and over-processing, dramatically raise the overall time and distance traveled.

1. Excessive Transportation

Moving clothes around the house through several processes leads to a very high transportation distance of 720 ft involved in the current process. The most significant causes are:

- Gathering dirty clothes from different rooms
- Going from the sorting area to the laundry room
- Moving wet clothes from the washer to the dryer
- Returning folded clothes to different rooms

These processes generate trips and movement that are both time-consuming and exhausting.

Positioning the laundry baskets centrally, keeping hampers in every bedroom, and putting the washer–dryer next to each other are ways that this movement can be reduced considerably.

2. Waiting (Delay Waste)

The washing machine cycle results in 25 minutes of waiting period where no value-added work is done. During the time that the machine is processing the load, the operator is also idle. This situation leads to non-productive time. While it is not possible to completely get rid of waiting, it could be a lot better organized by the use of phone timers or planning some other small tasks during this time.

3. Over-Processing and Repeated Handling

Some of the steps like sorting of clothes and checking for stains are done one after another, and this may cause the clothes to be handling repeatedly. Sorting and then checking for stains uplifts the operator to touch the laundry multiple times. Merging these two steps into one or using pre-sorted laundry hampers can lessen the duplication and reduce the total time required for the task.

4. Inefficient Layout

A bad home layout not only costs more material but also wastes more space. Sorting and folding

far from the laundry room are just two among the many places where clothing is stored in multiple rooms requiring extra trips. That is why it is so essential to have a professional organize the space as it will greatly speed up the process.

5. Absence of Standardization

The variability and time wasting are the resulting factors of the manual detergent dosing, manual sorting and inconsistent placement of clothes. The use of detergent pods, labeled drawers, and standard sorting categories can lessen the cognitive load and enhance the consistency.

Part 3: Proposed Section of Flow Process Chart

Event Description	Symbol					Time (min)	Distance (ft)	Method recommendation
Collect dirty laundry using the laundry hamper placed in each room	○	→	D	□	▽	8	60	The hampers placed to collect dirty laundry in each room reduces unnecessary trips
Sort the clothes and inspect stains simultaneously	○	→	D	□	▽	10	0	By using separate laundry baskets, it eliminates the need to sort again, and stains can be identified while loading the machine and treated accordingly
Using the wheeled hamper, carry the clothes to the laundry room	○	→	D	□	▽	3	80	Laundry baskets with wheels reduces transportation time and physical effort
Load the washer with detergent pods (pre-measured)	○	→	D	□	▽	3	5	Using detergent pods improve consistency and reduces the time needed to measure laundry detergent

Wash cycle is complete	○	→	D	□	▽	20	0	Using a timer to alert when the cycle is complete, thus eliminating the need to check completion
Transfer wet clothes to dryer (which is placed beside the washer)	○	→	D	□	▽	2	10	By placing the washer and dryer side by side, it reduces transportation time
Dry the clothes	○	→	D	□	▽	15	0	Using a dryer set on high efficiency mode reduces drying time and saves energy
Transfer the dried clothes to the folding station near the laundry room	○	→	D	□	▽	2	10	Setting up the folding station near the laundry room reduces transportation time
Fold the clothes at the dedicated station(surface/table)	○	→	D	□	▽	5	0	Using a surface that is ergonomically friendly increases speed and improves dexterity
Move the folded (and grouped) clothes to each room	○	→	D	□	▽	3	80	Grouping clothes belonging to individuals minimizes trips
Store clothes in closets/drawers	○	→	D	□	▽	4	60	A reorganized closet helps reduce storage time

Comparison Summary

Event	Present	Proposed	Savings
Operation	57	37	20 mins saved
Transportation	31	18	13 mins saved
Delay	45	20	25 mins saved

Inspection	5	0	5 mins saved
Total Time	138	75	63 mins saved
Transportation (ft)	670	305	365 ft saved
Estimated Cost	High	Medium	Reduced time and strain

The future-state process cuts down on inefficiencies by changing the physical layout, integrating manual tasks, and decreasing unnecessary movement. Significant reduction in transport distance comes from putting hampers in every room. The operation for sorting and stain-inspection has been merged into one which leads to no repeated handling. The use of a rolling laundry cart, detergent pods, and putting the washer and dryer side by side are all methods that help speed up the process and reduce the time taken.

A folding station that is specifically designed and placed close to the laundry area has the advantage of less motion waste and no more walking trips. Load of transportation is reduced even more by grouping the folded clothes according to the room they belong to. Better closet organization then acts as a catalyst in speeding up the storage process.

The cumulative effect of the changes is that walking distance is minimized, delays are reduced, and total cycle time is almost one hour less than before, thereby resulting in an improvement in both efficiency and ergonomics.